

Majorana zero modes in superconductor–semiconductor heterostructures

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Abstract | Realizing topological superconductivity and Majorana zero modes in the laboratory is a major goal in condensed-matter physics. In this Review, we survey the current status of this rapidly developing field, focusing on proposals for the realization of topological superconductivity in semiconductor–superconductor heterostructures. We examine materials science progress in growing InAs and InSb semiconductor nanowires and characterizing these systems. We then discuss the observation of robust signatures of Majorana zero modes in recent experiments, paying particular attention to zero-bias tunnelling conduction measurements and Coulomb blockade experiments. We also outline several next-generation experiments probing exotic properties of Majorana zero modes, including fusion rules and non-Abelian exchange statistics. Finally, we discuss prospects for implementing Majorana-based topological quantum computation.

The search for topological phases of matter has generated significant interest in physics, chemistry and materials science^{1–4}. The realization of topological superconductors supporting Majorana zero-energy modes (also known as Majorana zero modes, MZMs) is of fundamental scientific importance and would have profound technological applications for quantum information processing. MZMs correspond to exotic neutral excitations of a superconductor comprising an equal superposition of an electron and a hole. In this sense, a solid-state Majorana quasiparticle, which is its own antiparticle, is closely related to the Majorana fermions that were first introduced by Ettore Majorana in the context of elementary particle physics⁵. A neutral excitation in a superconductor has a special property owing to the inherent particle–hole symmetry of the material: it is bound to zero energy, so that there is no cost to occupy such a state. Therefore, the presence of a large number of localized MZMs leads to an extensive ground-state degeneracy, which can be exploited for topological quantum computing purposes^{6,7}. Theory^{8,9} predicts that MZMs localized at defects obey exotic exchange statistics, similar to that of non-Abelian anyons, which are neither fermions nor bosons. This is by far the most interesting property of MZMs, because it means that exchanging the position of MZMs corresponds to a non-trivial transformation within the degenerate ground-state manifold and represents a non-commutative operation, which does not depend on the method or details of its execution. Therefore, such

an operation is topologically protected (that is, it only depends on the exchange statistics of the quasiparticles) and may be used to implement quantum gates. These ideas are at the heart of the topological approach to quantum computing^{6,7}. Indeed, by exploiting topological materials, which by their nature minimize errors⁶, it might be possible to overcome the largest barrier to building scalable quantum computing: decoherence.

A number of platforms for realizing MZMs in the laboratory have been put forward^{10–18}. Most of them propose to engineer an appropriate model Hamiltonian at the interface between a conventional superconductor and topological insulators^{19–22}, semiconductors with strong spin–orbit coupling^{23–29} or magnetic atom chains^{30–45}. Among the most promising proposals are those based on proximitized nanowires^{25,26}; there is mounting experimental evidence for the presence of MZMs in these systems^{46–62}.

A paradigmatic model for a 1D topological superconductor involves spinless fermions that hop along a chain and experience proximity-induced *p*-wave superconducting pairing⁶³. Such a system supports an odd number of localized MZMs at the opposite ends of the chain. As explained above, these modes are neutral and thus a single MZM cannot accommodate a conventional (Dirac) fermion. At least two MZMs are needed to form a fermionic state with a well-defined occupation number. As a result, a 1D topological superconductor supports a non-local fermionic mode comprising two MZMs

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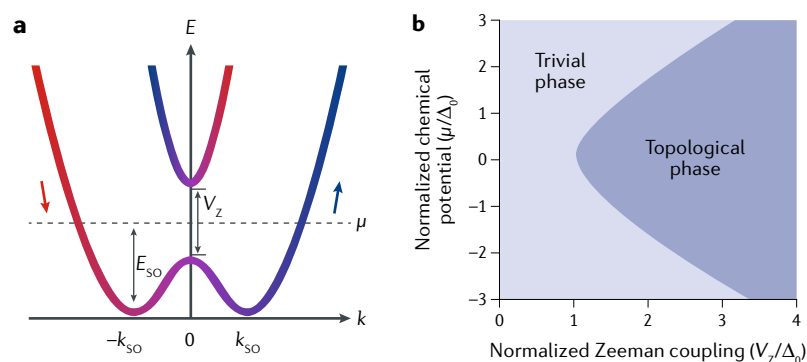


Fig. 1 | Energy spectrum and topological phase diagram of a semiconductor nanowire proximity-coupled to a superconductor. a | Energy spectrum as a function of the momentum k along the nanowire. Spin–orbit coupling shifts the parabolas describing the electron spectrum sideways by k_{so} and introduces a new energy scale, E_{so} . Zeeman coupling V_z , which is due to an external magnetic field perpendicular to the direction of the spin–orbit coupling, opens a gap at $k=0$ (REFS^{25,26}). The arrows indicate the approximate spin orientations for different momenta, the colours correlate to the spin orientation (red is spin down; blue is spin up), and the dashed line indicates the chemical potential μ . **b** | Topological quantum phase diagram. By changing the chemical potential μ in the nanowire or Zeeman splitting V_z , it is possible to drive the system into a topological phase. Here, Δ_0 is the proximity-induced pairing gap in the wire.

localized at opposite ends of the chain and separated by a distance that can be much larger than the superconducting coherence length. This non-local entanglement in a gapped system is the hallmark of topological superconductors and represents an instance of electron fractionalization⁶³. This argument relies on having an odd number of MZMs per wire end, because an even number of Majorana modes can pair up and locally form an Andreev state (which is a conventional fermionic state). Such a fermionic mode generically resides at a finite energy, and as a result, there is no non-local entanglement in the ground state. Thus, there is a profound difference between conventional gapped superconductors, which have a unique ground state with even fermion parity, and topological superconductors, which instead have a highly degenerate ground state owing to the presence of many MZMs^{9,63}.

Realizing the above model poses a materials science challenge. Spinless p -wave superconductivity is a key element for realizing separated MZMs. However, electrons in conventional materials have spin $1/2$; thus, the notion of a spinless superconductor does not seem immediately relevant to real physical systems. An elegant way to overcome this difficulty is to use spin–orbit materials, in which the spin and orbital degrees of freedom are correlated. Indeed, spinless superconductivity can effectively emerge in a semiconductor nanowire with strong spin–orbit interaction proximity-coupled to a conventional (s -wave) superconductor^{25,26}. The energy spectrum for a nanowire with spin–orbit coupling is shown in FIG. 1a. The spin–orbit interaction shifts the momentum parabolas sideways, and when a magnetic field is applied, the Zeeman term (which does not commute with the spin–orbit term) causes the opening of a gap in the spectrum at zero momentum (BOX 1). When the chemical potential μ is tuned to be in the gap, the spectrum has a single band crossing Fermi points; thus, in this regime, the system is effectively described by ‘spinless’ degrees of freedom

(FIG. 1a). However, owing to the spin–orbit-induced rotation of the spins at the opposite Fermi points, the proximity-induced s -wave interaction opens a pairing gap in the spectrum. The resulting state is closely related to a spinless p -wave superconductor (BOX 1).

One of the virtues of this model is that the proximitized nanowire can be driven into a topological phase by tuning the magnetic field or the chemical potential (FIG. 1b). The emergence of MZMs at a certain critical value of a control parameter is necessarily accompanied by the closing of the bulk gap^{9,63}, which corresponds to a topological quantum phase transition (that is, a quantum phase transition between topologically trivial and non-trivial states; FIG. 1b). The topological phase is stable with respect to small perturbations (such as disorder) as long as they do not cause the bulk gap in the spectrum to collapse. The ability to realize the topological phase depends on the effective spin–orbit energy E_{so} , the proximity-induced gap Δ_0 and the effective Zeeman energy V_z in the heterostructure. Therefore, when engineering materials to obtain MZMs, an obvious choice is to pick material components that individually have the required properties. So far, the heavy-element semiconductors InAs and InSb have received considerable attention owing to their strong spin–orbit coupling and large Landé g -factor, whereas Al and NbTiN have been used as the superconducting components. The relevant bulk properties of these materials are listed in TABLES 1, 2. The large g -factor of the semiconductor allows one to achieve the desired effect, opening a large effective Zeeman gap in the proximitized nanowire using in-plane magnetic fields with values below the critical field of the s -wave superconductor. High-quality interfaces between the two components lead to a significant hybridization of the semiconducting and metallic states, resulting in a large induced pairing gap. The interface should be smooth, because disorder scattering tends to suppress p -wave superconductivity and eventually leads to the collapse of the topological phase^{64–78}. All these requirements are important for the observation of MZMs and pose a challenging materials science task.

This Review is focused on the current state of the realization of MZMs in superconductor–semiconductor heterostructures; Majorana proposals based on topological insulators and chains of magnetic atoms have been covered in several other excellent reviews^{10,79–83}. We start with an overview of the recent materials science developments in the growth of high-mobility semiconductors and in the preparation of high-quality interfaces between superconductors and semiconductors, which make it possible to perform experiments to detect MZMs. We then summarize robust experimental signatures of MZMs and discuss the best evidence for the presence of MZMs in proximitized nanowires. Finally, we discuss the perspective for topological quantum computation with MZMs and proposals for experiments in this area.

Materials science aspects

Semiconductor growth. The fabrication of InAs and InSb semiconducting materials exhibiting 1D confinement has been achieved by two different methods. The first method involves the top-down (lithography-based) definition of 1D structures in 2D quantum wells that

Box 1 | A minimal model for 1D topological superconductor

The minimal model for a 1D topological superconductor is that of a semiconductor nanowire with strong spin–orbit coupling, proximity-coupled to a conventional (s-wave) superconductor^{25,26}. The effective low-energy Hamiltonian for such a system is given by

$$H = H_{\text{SM}} + H_{\text{P}}$$

where the term associated with the semiconductor is

$$H_{\text{SM}} = \sum_{\sigma, \sigma'} \int_0^L dx \psi_{\sigma}^{\dagger}(x) \left(-\frac{\hbar^2 \partial_x^2}{2m^*} - \mu + i\hbar\alpha \hat{\sigma}_y \partial_x + V_Z \hat{\sigma}_x \right) \psi_{\sigma'}(x)$$

and the term related to proximity coupling is

$$H_{\text{P}} = \int_0^L dx \left(\Delta_0 \psi_{\uparrow}^{\dagger}(x) \psi_{\downarrow}^{\dagger}(x) + h.c. \right)$$

Here, $\hbar$ is the Planck constant; m^* , μ and α are the effective mass, chemical potential and Rashba spin–orbit coupling in a semiconductor nanowire of length L , respectively; V_Z is the Zeeman splitting due to a magnetic field B_x applied along the nanowire, where $V_Z = g\mu_B B_x$ (g and μ_B are the Landé g -factor and Bohr magneton); $\hat{\sigma}$ are Pauli matrices; Δ_0 is the induced pairing gap; and $\psi_{\sigma}(x)$ is the wavefunction for an electron with spin σ . Typical material parameters for InAs and InSb semiconductors are provided in TABLES 1, 2, and values for the induced gap are given in TABLE 3. Although the semiconductor nanowire is coupled to an s-wave superconductor, the presence of spin–orbit coupling leads to spin-momentum locking and results in a mixed Cooper pair wavefunction consisting of singlet and triplet pairings¹⁸⁵. When the Zeeman term opens a large gap in the spectrum (that is, $V_Z > \sqrt{\mu^2 + \Delta_0^2}$), the singlet pairing component is suppressed, and the superconducting state that forms at the interface has an order parameter with p -wave symmetry. This state is adiabatically connected to a corresponding state of Kitaev's model⁶³ and supports Majorana zero modes (MZMs) at the opposite ends of the wire.

are grown by molecular-beam epitaxy. The quantum wells are grown close to the surface of the semiconductor, allowing for deposition or epitaxial growth of a superconducting material in close proximity to the well. The growth of 2D semiconductor materials has been studied for many years and is now well understood for many different material combinations. However, for semiconductor–superconductor heterostructures, new challenges emerge. To increase electron mobility in conventional 2D materials, an obvious strategy is to grow a thick insulating and, ideally, lattice-matched layer on top of the material hosting the 2D electron gas (2DEG) to protect the 2DEG from surface disorder. However, this also suppresses induced superconductivity, which relies on electron exchange between the semiconductor and superconductor. The solution is to grow a thinner and low-barrier material as an intermediate layer; the task is to find the barrier height and thickness that give the best combination of mobility and induced superconductivity⁸⁴. Moreover, because 1D confinement is defined by top-down methods using either mesa etching or selective etching of the superconductor, the need for controlled definition of the nanowires leads to additional requirements on the materials and etching processes. This is likely to be a topic of ongoing research over the coming years.

The other way to produce nanowires is to use the bottom-up vapour–liquid–solid (VLS) growth mechanism. This mechanism has been developed over the past 50 years and can be used to fabricate nanowires from a variety of semiconductor materials^{85–90}. InAs and InSb have shown promising behaviour in this fabrication scheme. Typically, a metal particle catalyst is used to

collect precursor material from the vapour phase and to establish local supersaturation. The catalyst becomes liquid when it forms a eutect with the precursor species, and when a critical supersaturation level is reached, solid material is precipitated at the liquid/substrate interface, resulting in layer-by-layer growth. This mechanism results in a crystalline nanowire with radius determined by the catalyst dimensions (FIG. 2a). Precursor material can directly impinge from the gas phase onto the catalyst surface, or it can first land on the substrate and subsequently diffuse towards the catalyst. While diffusing, the adatoms can be trapped, and when this happens on the side wall of the nanowire, it leads to radial growth. To fabricate 1D nanowires, it is important to suppress this effect. In the case of InSb nanowire growth, an additional problem is the low surface energy of Sb, which acts as a surfactant and therefore makes it difficult to initiate nanowire growth, because the wetting layer tends to form a 2D layer that blocks nanowire growth. One way to overcome this problem and to reduce radial growth is by growing a heterostructure. For example, an InP/InAs stem wire can be used to grow an InSb wire^{86–90} (FIG. 2b–d). These wires can be a few micrometres long and have diameters of 80–100 nm. The crystal structures of nanowires grown by VLS are relatively easily investigated by transmission electron microscopy (TEM). The defect-free cubic (zinc blende) structure of an InSb nanowire grown in the [111]B direction is shown in FIG. 2a–d. By contrast, defect-free InAs nanowires are typically grown in the hexagonal (wurtzite) crystal phase with a [0001]B axis. Nanowires have a strong tendency to grow in one of these two directions, and this tendency can be used to direct the orientation of the wires. Uniform

Table 1 | Bulk properties of semiconductors^{46,186,187}

Bulk properties	InAs	InSb
g-Factor	8–15	40–50
Effective mass m^* , m_e	0.023	0.014
Spin–orbit energy, meV	0.05–1	0.05–1
Spin–orbit coupling α , eV Å	0.2–0.8	0.2–1
Spin–orbit length, nm	180–40	230–50

Rashba spin–orbit coupling strength was measured in nanoscale structures^{84,107}. m_e is the electron mass; $E_{\text{SO}} = \frac{m^* \alpha^2}{2 \hbar^2}$ is the spin–orbit energy; $\lambda_{\text{SO}} \equiv k_{\text{SO}}^{-1} = \frac{\hbar^2}{am^*}$ is the spin–orbit length.

wires were used for the first-generation Majorana experiments⁴⁶, and their electronic properties are discussed in a later section.

For next-generation Majorana experiments, more complex nanowire structures are required, for instance, trijunctions (T-junctions), as explained below. An important requirement is that the junction between the branches has a small reflection amplitude for an incoming electron. While such structures can be processed ex situ using top-down lithography in 2D materials, VLS-grown structures can be fabricated in situ using the crystallography of the substrate^{91–97}. For example, instead of a (111) substrate, it is possible to use a (001) substrate in which trenches have been etched to expose the (111)B facets; this turns the growth direction away from the vertical. By using electron-beam lithography, the catalysts are defined on the sloped facets (FIG. 2e), and nanowires grow perpendicular to these facets in two equivalent $\langle 111 \rangle$ directions (FIG. 2f). After growth of the InP stem, the InSb segment is grown. Because all wires are epitaxially connected to the substrate, they form a single crystal when they merge. For the merging process, there are two important aspects to be considered. First, the position of the catalysts should be chosen so that there is the appropriate offset (indicated as the junction offset in FIG. 2e). If the offset is too small, the catalyst particles merge together on one of the nanowire side facets, and a nanowire bridge forms. If the offset is too large, the wires miss each other. An optimum offset is equal to the wire radius. Second, the merging process itself is governed by radial growth, which, as mentioned above, takes place in competition with axial growth⁹³. Radial growth results in an epitaxial shell and is responsible for the merging of two nearby wires to form crosses or hashtags (FIG. 2g,h). With this method, a predefined number of wires can be connected to form a network, and the dimensions and symmetry of the resulting structures can be tuned by changing the spacing of the trenches (FIG. 2e). The crossed wires have a single-crystal structure, as revealed by TEM studies^{93,97,98}.

Superconducting interfaces. First-generation Majorana experiments^{46,48–51} showed a large subgap conductance, indicating the presence of unintended subgap states, which are detrimental to topological quantum computation because they degrade topological protection. A likely source of the so-called soft gap, or continuum of subgap states, is disorder at the

semiconductor–superconductor interface⁷³. For devices based on Majorana quasiparticles, the native oxide that forms when the semiconducting wires are exposed to ambient air must be removed before deposition of the superconductive material. There has been significant progress in improving these interfaces, which are obtained by gently etching the oxide and optimizing the deposition of the superconductor^{53,59,99}.

Another important step forward was taken by epitaxially growing thin Al films on pristine facets of the nanowires without breaking the vacuum⁹⁹. This method of in situ superconductor growth yields further significant suppression of the subgap density of states¹⁰⁰. Subsequently, high-quality epitaxial superconductor–semiconductor interfaces have been realized in other heterostructures^{60,84,98,101,102}. Typical values of the proximity-induced superconducting gap in different semiconductor–superconductor heterostructures are listed in TABLE 3. In situ shadow deposition, in which semiconductor nanowires are used as masks for deposition of the superconductor, can be used to define junctions in the superconductor without additional lithography and etching steps, avoiding detrimental disorder from top-down processing^{97,98}.

The basic requirements for the Al film are that it should be thin enough to withstand high parallel magnetic fields without being driven to the normal state and that it should have a uniform morphology along the nanowire, which ensures translational invariance and avoids disorder-induced subgap states. The equilibrium shape of Al on a large, rigid InAs surface is not a continuous film but a dewetted droplet with a contact angle of about 60–80°. However, if the characteristic migration length of the Al adatoms is much shorter than the nominal thickness of the deposited Al, the Al can be kinetically locked to a continuous film. The key to obtaining a short adatom migration length is to lower the temperature sufficiently below room temperature. The specific temperature needed is highly dependent on the type of interface and type of structure. Moreover, because the native oxide stabilizes the film morphology, it is important to take the sample out of vacuum before it reaches the dewetting temperature, which for Al on InAs is typically below room temperature (the specific transition temperature depends on the type of interface). Unlike InAs growth, in which the crystal structure and morphology are locked once the wire is formed, the Al phase can easily recrystallize through grain growth transitions, which, at the right temperatures, move the grain boundaries while keeping the film continuous.

The grain growth process is driven exponentially by minimization of thermodynamic excesses such as strain and interfacial free energies. The measure of the

Table 2 | Bulk properties of superconductors^{46,187,188}

Bulk properties	Al	NbTiN
Superconducting gap Δ	0.2 meV	3 meV
Critical field B_c	10 mT	10 T
Critical temperature T_c	1.2 K	15 K

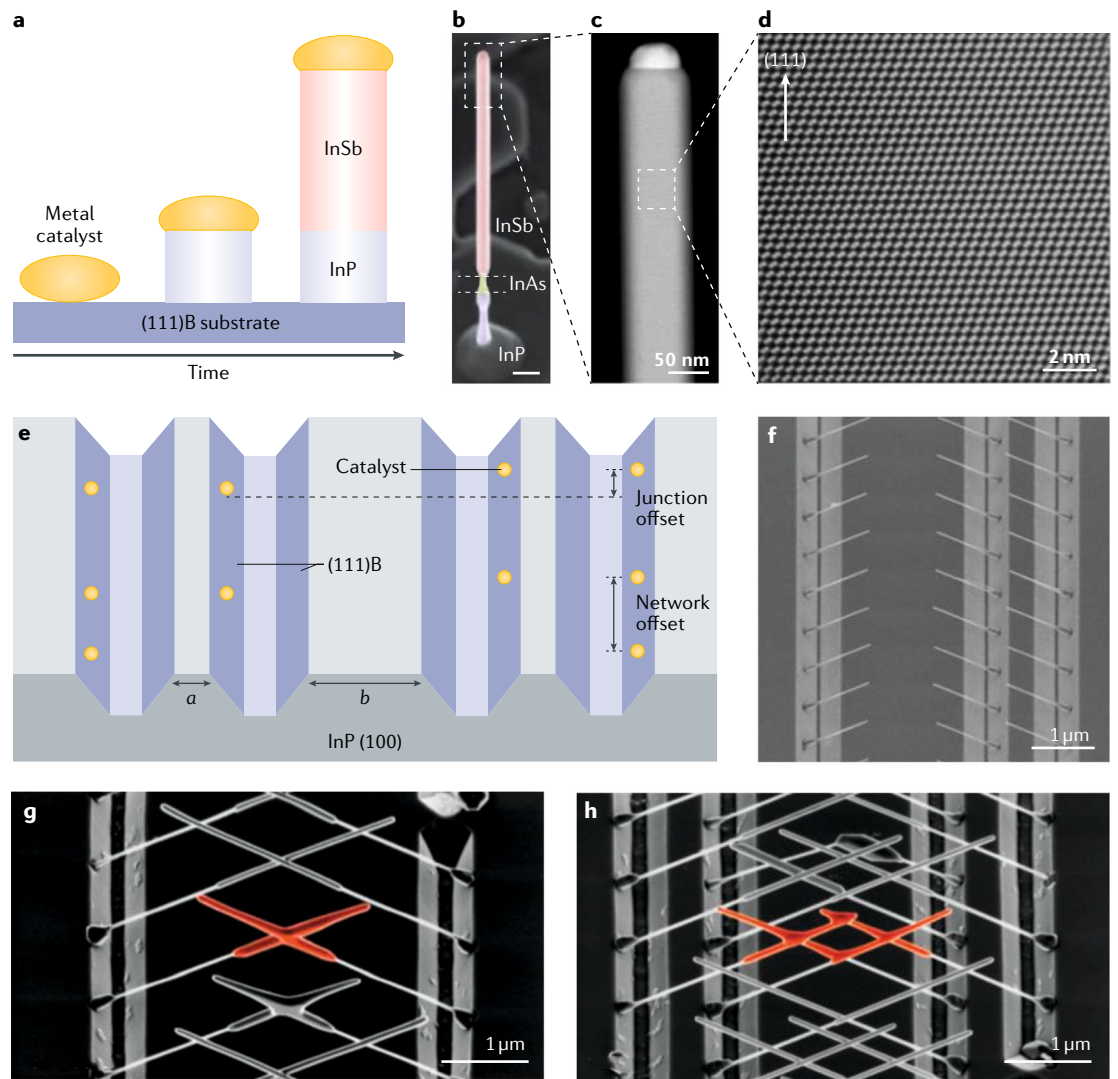


Fig. 2 | Growth of bottom-up nanowires. Schematic of the vapour-liquid-solid nanowire growth mechanism for an InP/InSb nanowire grown along the $\langle 111 \rangle_B$ crystal direction (part **a**), accompanied by a scanning electron microscopy (SEM) image of the nanowire (part **b**), a transmission electron microscopy (TEM) image showing the defect-free nature of the InSb wire (part **c**) and a high-resolution TEM image demonstrating the $\langle 111 \rangle$ growth direction (part **d**). Schematic of a generic method to fabricate a nanowire network with a predefined number of wires (part **e**). Trenches are etched from a (100)-oriented InP substrate so that (111)B facets are exposed; the spacings, indicated as a and b , can be used to tune the dimensions and symmetry of the resulting network. Electron-beam lithography is used to define catalysts on the (111)B facets in controlled positions. SEM images of an InP nanowire grown from the catalysts on the sloped facets (part **f**), InP/InSb nanowires forming a cross (part **g**) and InSb nanowire hashtags (part **h**). On one cross and one hashtag, InSb is indicated in false-colour red. Scale bars are $1 \mu\text{m}$. Panels **b–d** are adapted with permission from REF.⁹⁰, American Chemical Society. Panels **e–h** are adapted from REF.⁹⁸, Macmillan Publishers Limited.

thermodynamic driving force is the excess chemical potential. In the limit of low film thickness, for a given grain, this can be divided into four terms corresponding to the surface, superconductor–semiconductor interface, grain boundary and strain contributions^{84,99}:

$$\mu\alpha \frac{\gamma_{\text{surface}}}{h} + \frac{\gamma_{\text{SC-SM}}}{h} + \frac{\gamma_{\text{GB}}}{R} + \frac{S\varepsilon^2}{1-\nu} \quad (1)$$

Here, γ_{surface} is the energy density of the surface, $\gamma_{\text{SC-SM}}$ is the energy density of the superconductor–semiconductor interface, γ_{GB} is the energy density of the grain boundary, S is the shear modulus, ε is the strain, h and

ν are the thickness and Poisson ratio of the thin-film material and R is the mean in-plane radius of curvature of the grain boundary.

The epitaxial growth of Al on the facets of free-standing InAs nanowires is a more complex and dynamical process than on a planar InAs substrate and was described for a 2D platform in REF.⁸⁴. In the single interface case, the dominating term for thin uniform layers (when $h \ll R$) is the surface term, because $\gamma_{\text{surface}} > \gamma_{\text{SC-SM}}$. This fixes the out-of-plane orientation to the lowest energy facet, which is the (111) facet (FIG. 3a). The in-plane orientation appears to be driven primarily by the superconductor–semiconductor interfacial bicrystal symmetries, which are governed by

Table 3 | This maximum values of the proximity-induced superconducting gap Δ_0 ^{53,55,58}

Superconducting gap	InSb/NbTiN	InSb/Al	InAs/Al
Δ , meV	3	0.2–0.3	0.2–0.3
Maximum of Δ_0 , meV	1	0.2	0.2

The superconducting gap of Al depends on the film thickness¹⁰⁵. Note that Δ_0 in Al heterostructures is close to the bulk superconducting gap, indicating a strong tunnelling regime between the semiconductor and superconductor.

the second term in equation 1. As discussed in REF.⁸⁴, only two degenerate in-plane grain orientations of Al on planar InAs (100) are observed, indicating that these orientations are local minima of the interface energy subject to the constraint that the out-of-plane orientation is [111]. However, the size of the grains is driven by the minimization of the grain boundary energy, which is governed by the third term in equation 1. The final term in equation 1 (which is associated with the strain) plays an important role if the particular bicrystal lattice match at the superconductor–semiconductor interface has substantial mismatch but a significant minimum in the respective interfacial energy for the particular match. However, for a critical value of h of around 5 nm, the Al phase on a planar InAs substrate was measured to be fully relaxed⁸⁴, which means that the critical thickness for relaxation is presumably much less and therefore does not play an important role in the growth kinetics for this material combination. For free-standing nanowires with Al growing on multiple facets, the grain boundary energies play a dominant role in the overall orientation, especially as the film gets thicker. With thicker layers, the Al recrystallizes from a [111] out-of-plane orientation, as in the case of a single plane, to an orientation that lowers the total grain boundary energy. In other words, the Al orientation changes so that it matches the symmetry of the nanowires and minimizes the incoherent grain boundary excesses across adjacent facets (FIG. 3b). The surface of the Al becomes more uneven because it still has a tendency to form non-planar (111) facets. However, thin Al films (needed to withstand high magnetic fields) can stay extremely flat and uniform (FIG. 3c).

Characterization of the heterostructures. Together with strong spin–orbit coupling and induced superconductivity, a key requirement for MZMs is quasi-ballistic electron transport with a controlled odd number of occupied modes. In the ballistic regime, the motion of 1D-confined electrons is restricted to discrete energy bands, which results in quantized conductance plateaux. The observation of quantized conductance therefore provides direct evidence for quasi-ballistic transport in the nanowires and allows control of the mode occupation. However, this is difficult to achieve in nanowires because, in a quasi-1D geometry, impurities and scattering at the nanowire/contact interface result in an increased probability of reflection, which obscures the observation of quantized conductance. To create highly transparent contacts, InSb wires can be exposed to a S etch to remove the oxide layer before Cr/Au contact deposition. The resulting nanowire device consists of bottom gates on which a hexagonal boron nitride flake is deposited as a high-quality dielectric¹⁰³ (FIG. 4a, inset). The samples are mounted in a dilution refrigerator with a base temperature of 15 mK. The differential conductance G

is measured as a function of gate voltage V_{gate} and bias voltage V_{bias} (FIG. 4a). At zero bias voltage, an extended plateau is visible at $G_0 = 2e^2/h$, demonstrating ballistic transport through the lowest nanowire sub-band, with an extracted mean free path $l \approx 300$ nm. An additional small plateau at $2G_0$ is observed, indicating that the device has a small but finite energy splitting between the second and third sub-bands. At finite bias voltage, the two-terminal conductance is quantized in integer values of G_0 only if both the source and drain chemical potentials lie within the same sub-band. This creates diamond-shaped regions of constant conductance, indicated by black dotted lines in FIG. 4a. At the tip of the diamond, the two dotted lines cross when V_{bias} is equal to the sub-band energy spacing ΔE_{sub} , which is estimated to be $\Delta E_{\text{sub}} \approx 16$ meV.

Phase-coherent transport is a basic requirement for the interferometric readout schemes of the topological data, as described below. To study the transport properties of nanowire networks, the structures are transferred from the growth chip to the desired chip for device fabrication¹⁰⁴ using a nanomanipulator in a scanning electron microscope (SEM). An SEM image of the contacted nanowire hashtag structure is shown in the inset of FIG. 4b. In the magnetoconductance of the device (FIG. 4b), Aharonov–Bohm oscillations can be clearly seen. As expected, the period of the oscillations scales with the area of the hashtag device, demonstrating phase-coherent transport through the device. From temperature-dependent measurements, a phase-coherence length of a few micrometres has been estimated⁹⁸.

As mentioned above, it is important to eliminate unintended subgap states in order to achieve a hard superconducting gap, which protects against thermal quasiparticle excitations. The density of states in the proximitized nanowire can be probed with tunnelling spectroscopy by measuring the differential conductance as a function of the source–drain voltage. A comparison between Al conventionally evaporated and epitaxially grown on InAs nanowires is shown in FIG. 4c. With epitaxially grown Al, the ratio of the above-gap to subgap conductance is about 1:100. Interestingly, for a thin Al layer grown on two facets of the nanowire, the superconducting gap is hard (FIG. 4d). Furthermore, the size of the gap increases with decreasing Al thickness down to several nanometres¹⁰⁵. The devices with a thin Al layer also withstand high magnetic fields of up to ~ 3 T (REF.⁵⁷) (FIG. 4d), which satisfies an important requirement for reaching the topological phase transition^{25,26}, which occurs when Zeeman splitting in the semiconductor is sufficiently large (BOX 1) and the magnetic field does not suppress superconductivity in the Al shell.

Another important material requirement is strong spin–orbit coupling in the semiconductor. Spin–orbit

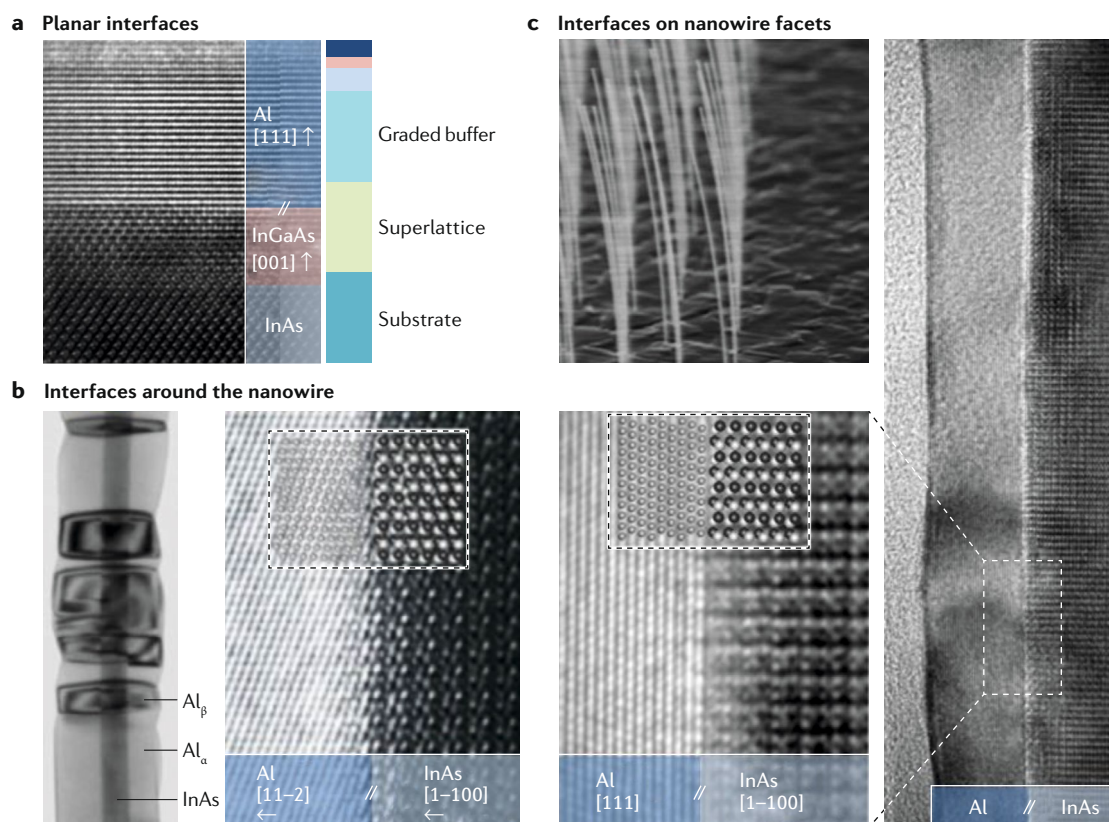


Fig. 3 | High-symmetry epitaxial Al/InAs interfaces. **a** | Epitaxial Al grown on an InGaAs barrier layer above a planar InAs layer (left). The dominant Al face-centred-cubic crystal orientation at low temperatures is [111] out of plane. Unlike the nanowire crystals, the planar epitaxial heterostructure shows better performance with an InGaAs barrier layer between the Al and InAs. A typical example of a layer stack grown on a lattice-mismatched substrate, such as InP (right). The superlattice traps the impurities, and the graded buffer makes the transition to the lattice constant of the active region⁸⁴. **b** | Thick Al layers on the nanowire facets typically lead to a gradual transition to a [11-2] out-of-plane crystal reconstruction, which allows for large single-crystal segments that follow the overall symmetry of the InAs core (left). The dark and bright contrasts, denoted by Al_{α} and Al_{β} , are two degenerate twin orientations. The resulting Al can form a complete coherent InAs/Al bicrystal phase with a near-lattice-matched 3:2 ratio (right). **c** | Scanning electron microscopy (SEM) image of an array of InAs/Al nanowires with Al sitting on two adjacent facets (upper left). The nanowires bend towards the Al-covered side. A thin, uniform Al phase is obtained with [111] out-of-plane crystal orientation (right), which is also shown under high-resolution magnification (bottom left). Panel **a** is adapted with permission from REF.⁸⁴, American Physical Society. Panels **b** and **c** are adapted from REF.⁹⁹, Macmillan Publishers Limited.

coupling is difficult to measure, especially in ballistic quasi-1D nanowires. A standard approach is to quench weak anti-localization with a small magnetic field; the field strength required to quench the zero-field conductance peak associated with quantum-suppressed backscattering yields an estimate for the strength of spin-orbit coupling within a rather complicated theoretical model¹⁰⁶ that includes different scattering processes. The phase-coherence length and strength of the Rashba spin-orbit coupling can be extracted from a fit to a weak anti-localization peak from a 2DEG⁸⁴ (FIG. 4e). The values measured in this InAs 2DEG are comparable to those measured in InAs and InSb nanowires^{84,107}. In addition, as can be done for nanowires, the density of states in 2DEGs can be measured using a tunnel probe, which in this case is a quantum point contact formed by top gates (FIG. 4f, inset). The gate-controlled quantum point contact can be tuned from a high-transparency regime (that is, a single conducting channel showing $4e^2/h$ two-terminal

conductance) to the weak tunnelling regime, in which the conductance is proportional to the local density of states (FIG. 4f).

Experimental signatures of Majorana zero modes

Even when all material requirements are satisfied, questions remain regarding how MZMs can be identified experimentally and how ambiguous various signatures are. Even when the system is in a trivial phase, without well-separated MZMs, fine-tuning some system parameters can lead to experimental observations with signatures that resemble those expected for MZMs. A system in the topological phase, however, shows these signatures in a robust way, that is, they persist under a moderate evolution of system parameters. It is this robustness that is the most characteristic trait of topological phases and ultimately what makes these states useful for quantum information processing. Below, we discuss a set of experiments showing Majorana signatures; the existence of more than one experiment with consistent results

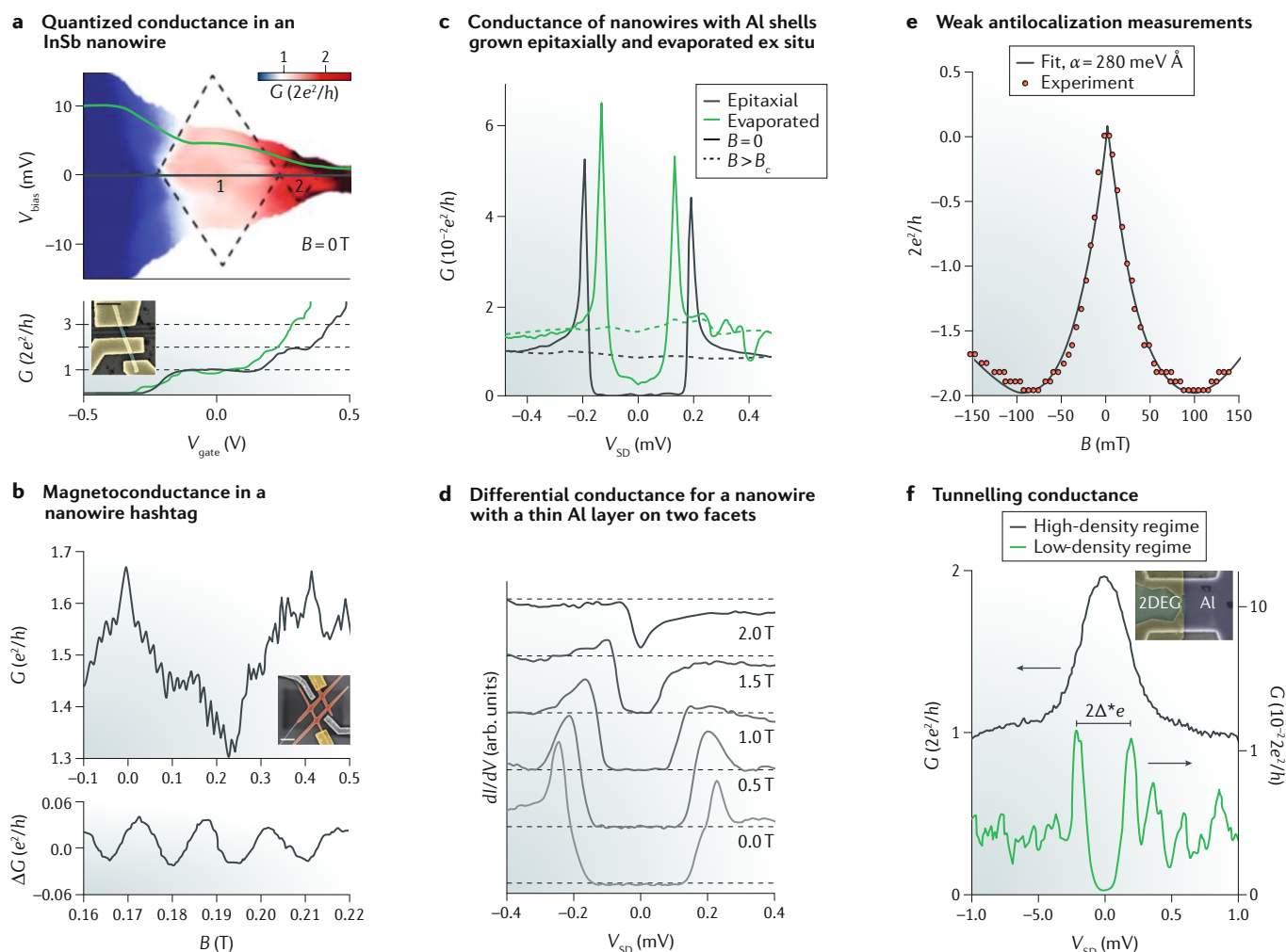


Fig. 4 | Characterization of superconductor–semiconductor heterostructures. **a** | Quantized conductance G through an InSb wire as a function of gate voltage V_{gate} and bias voltage V_{bias} (upper panel) at zero magnetic field B , and G as a function of V_{gate} at the cut-through lines (green and dark grey) depicted in the upper panel (lower panel). Inset: scanning electron microscopy (SEM) image of an InSb nanowire with ohmic contacts. **b** | G (and ΔG) as a function of B . Aharonov–Bohm oscillations in an InSb nanowire hashtag demonstrate phase-coherent transport. Inset: SEM image of an InSb nanowire hashtag. **c** | G as a function of the source–drain voltage V_{SD} of an epitaxial full-shell InAs/Al nanowire compared with that of a nanowire with an Al shell evaporated ex situ. For each wire, the measurements are reported for $B = 0$ and B higher than the critical field B_c . **d** | Tunnelling spectra show a hard induced gap under magnetic fields up to 1.5 T in an InAs nanowire with a two-facet epitaxial Al film. The wire is depleted by a gate and shows no subgap states in this regime. **e** | Weak anti-localization measurements on a Hall bar with a 5 nm InGaAs barrier and Al selectively etched away. The change in conductivity with respect to its value at zero field, $\delta\sigma$, allows extraction of the magnitude of the strength of the spin–orbit coupling in the semiconductor, α , which is equal to 280 meV Å. **f** | Tunnelling conductance in the high-density (black line) and low-density (green line) regimes. The tunnelling conductance in the low-density regime provides an estimate for the induced superconducting gap. The system is driven between the two regimes by changing the voltage applied to the quantum point contact (V_{QPC}). Inset: SEM image of the device. 2DEG, 2D electron gas; dI/dV , differential conductance. Panel **a** is adapted with permission from REF.¹⁰³, American Chemical Society. Panel **b** is adapted from REF.⁹⁸, Macmillan Publishers Limited. Panel **c** is adapted from REF.¹⁰⁰, Macmillan Publishers Limited. Panel **d** is adapted with permission from REF.⁵⁵, AAAS. Panel **e** is adapted with permission from REF.⁸⁴, American Physical Society. Panel **f** is adapted from REF.¹⁸⁹, Macmillan Publishers Limited.

supports the case that proximitized nanowires can be driven into a topological superconducting phase. We start by introducing the variety of experiments that have been performed, before discussing in detail zero-bias tunnelling and Coulomb blockade experiments.

The simplest experiments involve tunnelling conductance measurement of the local density of states at the

nanowire ends^{66,108–115} (FIG. 5a). The dependence of the differential tunnelling conductance on the magnetic field is shown in FIG. 5b. The characteristic zero-bias peak in tunnelling conductance appears at a finite field, which is a signature of MZMs. The height of the zero-bias peak is predicted to be $2e^2/h$ at zero temperature^{110,116,117}. In addition to zero-bias tunnelling spectroscopy, a

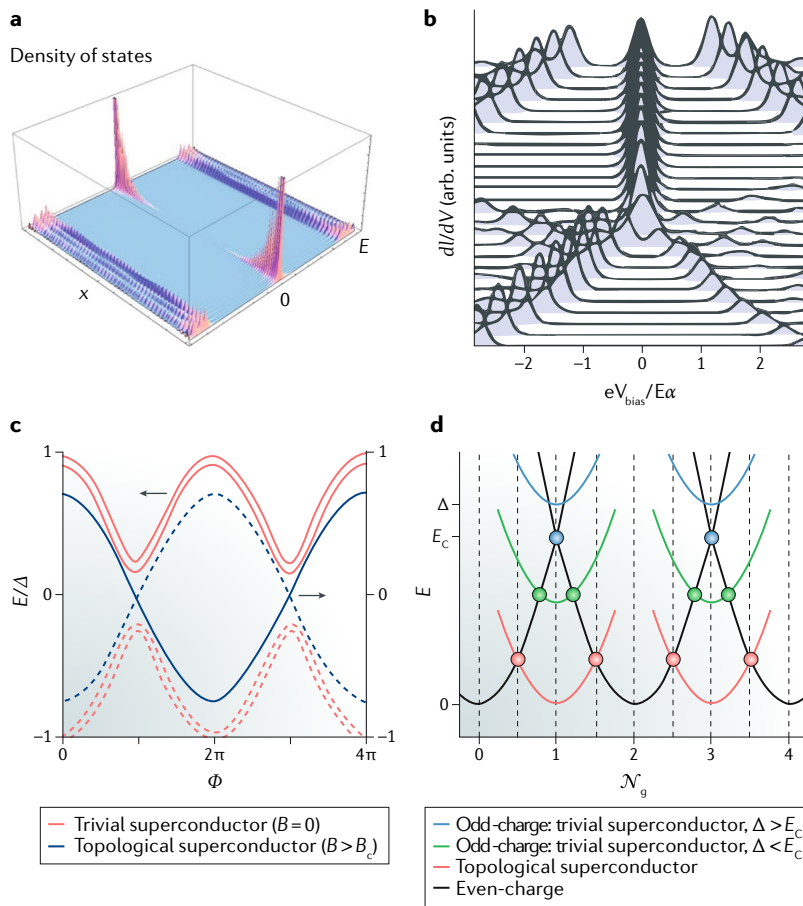


Fig. 5 | Signatures of Majorana zero modes. **a** | Simulated local density of states as a function of energy E and coordinate x along the nanowire for a proximitized nanowire in a topological phase with localized Majorana zero-energy states at its ends. **b** | Differential tunnelling conductance (dI/dV) as a function of applied bias voltage, V_{bias} . Each line cut corresponds to a different in-plane magnetic field B . At a critical value of B , a Majorana-induced zero-bias peak emerges. This feature is stuck to zero bias for a considerable range of values of the magnetic field. Here, $E\alpha$ is the spin-orbit coupling energy. **c** | Energy spectrum of Andreev bound states in a superconductor–nanowire–superconductor junction as a function of superconductor phase difference Φ across the junction. Red and blue lines correspond to the energy spectrum for trivial ($B=0$) and topological ($B>B_c$) superconductors, respectively, where B_c is the critical field. In the trivial phase, there is an avoided level crossing at $\Phi=\pi, 3\pi$ owing to the double degeneracy of the spectrum required by time-reversal symmetry, whereas in the topological state, this crossing is protected by particle–hole symmetry⁶³. Dashed and solid blue energy states have different fermion parity. Dashed red lines correspond to quasi-holes, and solid red lines refer to quasiparticles. **d** | Ground-state energy E of a mesoscopic proximitized nanowire as a function of the dimensionless gate voltage N_g . Black lines correspond to the ground-state energy for even-charge N_g . Blue and green lines correspond to the odd-charge states of a conventional (trivial) superconductor with $\Delta>E_c$ and $\Delta<E_c$, respectively (Δ is the quasiparticle gap). The red line represents the ground-state energy of an odd-charge state in a topological superconductor, in which case even-charge and odd-charge states are (almost) degenerate owing to the presence of Majorana zero-energy modes. Peaks in transport occur at the points indicated by the filled circles. Panels **a** and **b** are adapted with permission from REF.⁶⁶, American Physical Society. Panel **d** is adapted from REF.⁵², Macmillan Publishers Limited.

number of current correlation measurements probing the presence of MZMs have been proposed^{109,110,118–123}.

Another peculiar feature of MZMs is that they lead to the fractional Josephson effect. This effect corresponds to coherent tunnelling of single electrons between two topological superconductors forming a

Josephson junction. As a function of magnetic field, the spectrum of Andreev states is predicted to change from being 2π periodic in units of superconductor flux quantum to 4π periodic (FIG. 5c); 2π periodicity corresponds to the tunnelling of Cooper pairs, whereas 4π periodicity develops because of coherent single-electron tunnelling processes. The characteristic crossing of two Andreev states at zero energy is protected by fermion parity in the junction⁶³. When parity is conserved, there is a true level crossing in the energy spectrum, leading to 4π periodicity of the supercurrent. In practice, finite-size effects lead to a small avoided level crossing, and a dynamical change in the fermion parity may also occur owing to unpaired electrons in the system. Therefore, supercurrent measurements should be performed at sufficiently high frequency that these effects are avoided^{25,63,124–128}.

Another signature of MZMs appears when a semiconductor nanowire is proximitized by a floating superconductor (rather than a grounded superconductor as discussed so far), such that the overall system has a finite charging energy^{129–135}. The corresponding Hamiltonian reads

$$H_C = E_C (\hat{N} - N_g)^2 \quad (2)$$

where $\hat{N}$ is the total number of electrons in the mesoscopic system (island); $E_C = e^2 / 2C_\Sigma$ is the charging energy, where e and C_Σ are the electron charge and the total capacitance of the nanowire and its superconducting shell, respectively; and $N_g = C_g V_g / e$ is the dimensionless gate voltage, where C_g and V_g are the gate capacitance and gate voltage, respectively.

In a conventional (s-wave) superconductor with an even number of particles, all electrons are paired in the ground state. Adding one more electron costs electrostatic energy as well as pairing energy $\Delta(B)$. Therefore, if $\Delta(B) > E_C$, the ground-state energy of the proximitized nanowire corresponds to an even-charge sector with $2e$ -periodic dependence on N_g (FIG. 5d). The presence of a magnetic field leads to the suppression of the quasiparticle gap $\Delta(B)$, and the parity of the ground state can have even-charge and odd-charge sectors, but the ground state remains $2e$ periodic as long as $B < B_c$. In the topological phase ($B > B_c$), the periodicity of the ground-state energy changes owing to the presence of MZMs and becomes $1e$ periodic¹²⁹ (FIG. 5d). This periodicity, in combination with the presence of elastic transport through a Coulomb-blockaded (Majorana) island, provides evidence for MZMs in the proximitized nanowires¹³³. By contrast, transport through a metallic island (that is, no MZMs) is dominated by inelastic processes¹³⁶.

An interesting situation arises when a Coulomb-blockaded island contains several pairs of MZMs. When such an island is coupled to several normal leads, Majorana-induced ground-state degeneracy leads to an exotic multichannel Kondo effect^{137–139}.

All these properties characterize a topological phase and, as a result, persist in a finite parameter range. It may be possible to explain each of these physical observations

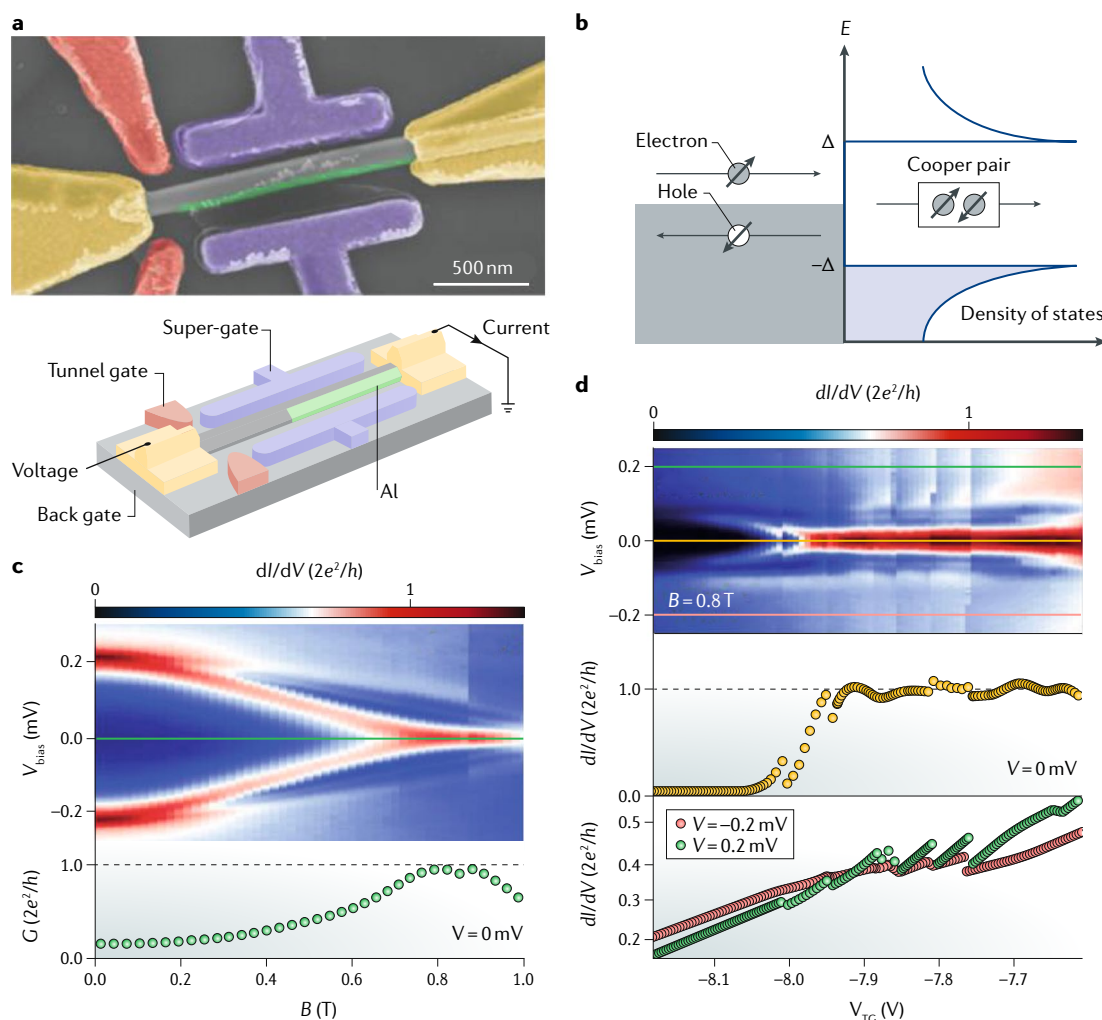


Fig. 6 | Tunnelling conductance in InSb/Al hybrid structures. **a** | The layout of the device. Upper panel: false-colour electron micrograph. Lower panel: device schematic. Here, the side gates and contacts are Cr/Au (10 nm/100 nm), and the Al shell thickness is 10 nm. The substrate is p-doped Si, which acts as a global back gate. The barrier between the normal lead and the proximitized nanowire can be controlled by the two tunnel gate voltages. **b** | Schematic of Andreev reflection processes at a normal metal–superconductor interface. **c** | Upper panel: tunnelling conductance as a function of the in-plane magnetic field (up to 1 T) and bias voltage V_{bias} . The induced gap in this device is $\Delta_0 = 0.2$ meV; the zero-bias peak is visible between 0.7 and 1 T. Lower panel: zero-bias conductance line cut as a function of magnetic field. In this experiment, the temperature is $T = 20$ mK. **d** | Dependence of the tunnelling conductance on the tunnel gate voltage V_{TG} and V_{bias} at a fixed $B = 0.8$ T. Lower panels show horizontal line cuts at zero and finite bias. The zero-bias conductance is robust in the large parameter regime and remains close to the quantized value of $2e^2/h$, which distinguishes this case from that corresponding to a trivial Andreev bound state, in which the height of the conductance peak ranges between 0 and $4e^2/h$. Δ , gap energy; dI/dV , differential conductance; DOS, density of states; E , energy. Panels **a**, **c** and **d** are reproduced from REF.⁵⁸, Macmillan Publishers Limited.

separately within certain non-MZM scenarios in which the system parameters are fine-tuned. However, as discussed below, a large body of experimental data accumulated in different physical systems makes it possible to rule out many of the false-positive interpretations in favour of an MZM scenario.

Zero-bias tunnelling conductance measurements.

Tunnelling spectroscopy is a conceptually simple method of measuring the local density of states and is well suited to detect reconstruction of the energy spectrum across a topological phase transition. Given that MZMs are localized at the ends of the nanowire,

the differential conductance measurement is expected to exhibit a robust zero-bias tunnelling conductance peak in the topological phase. A typical device layout for tunnelling spectroscopy measurement is shown in FIG. 6a. Low-bias transport in the normal metal–superconductor junction is predominantly determined by Andreev processes, in which an incident electron is reflected as a hole, and a Cooper pair is added to the superconducting condensate (FIG. 6b). In the non-topological phase ($B < B_c$), these processes are suppressed, and the junction is weakly transparent. By contrast, the transmission probability approaches one in the topological phase^{110–115,140,141}. The presence of

MZMs leads to a resonant Andreev reflection, and as a result, the differential tunnelling conductance acquires a universal value of $2e^2/h$ at temperatures that are low compared with tunnel broadening^{110,116,117}.

Shortly after theoretical proposals for proximitized nanowires were published^{25,26}, the emergence of a zero-bias peak in a finite magnetic field was observed⁴⁶. Similar results were published in several subsequent studies^{48–51,54} on InSb and InAs nanowires. First-generation experiments^{46,48–51} reported the appearance of a zero-bias peak together with a closing of the bulk gap. The zero-bias feature persisted over a substantial range of magnetic fields and gate voltages and disappeared with suppression of the superconductivity. Furthermore, the zero-bias peak disappeared when the in-plane magnetic field was aligned transverse to the wire axis, indicating that the effect depended on the interplay of Zeeman and spin–orbit couplings. All these observations are consistent with the presence of MZMs. However, first-generation experiments also revealed a large subgap density of states due to interfacial disorder, which coexisted with the zero-bias feature. It is challenging to separate the two contributions, and various alternative interpretations have been proposed involving disorder-induced weak anti-localization^{68,125,142}, the formation of quantum dots^{143,144} or the presence of smooth confining potentials¹⁴⁵.

More recent experiments with improved deposited InSb/NbTiN interfaces^{53,54}, epitaxial InAs/Al nanowires⁵⁵ and 2D InAs/Al heterostructures^{56,57}, and epitaxial InAs/Al and InSb/Al interfaces^{58,60} showed a much lower subgap density of states (FIG. 6c) as well as ballistic transport properties that make it possible to rule out the scenario of a disorder-induced zero-bias conductance peak. It is the materials science progress reviewed in this work that has enabled significant improvement in electric transport and allowed investigation of subtle transport properties, such as quantization of the zero-bias tunnelling conductance. For example, early experiments^{46,48–51} reported a zero-bias peak much smaller than the predicted value of $2e^2/h$, but recent improvements in sample quality have led to an increase in the intensity of the zero-bias conductance peak, and a G value on the order of e^2/h was observed in InAs/Al and InSb/Al devices^{58,146}. Improved InSb/NbTiN samples have been used to map the topological phase diagram through the dependence of the zero-bias peak on the chemical potential and magnetic field^{53,54}. These measurements contradict many false-positive hypotheses. However, Andreev bound states may exhibit similar signatures^{141,145}. To distinguish between local Andreev bound states and delocalized MZMs, it is necessary to carry out a detailed analysis of how the observed signal depends on the temperature and tunnel coupling^{57,128} (which was recently reported experimentally in 1D and 2D heterostructure devices^{57,58}) or to consider more complex geometries, for instance, a quantum dot coupled to the end of a nanowire^{143,144,147–152}, which allows measurement of the non-locality of MZMs⁶¹. Tunnelling conductance due to local Andreev states depends on the tunnel barrier height, whereas Majorana-induced zero-bias peak is robust with respect to the tunnel

coupling and in fact exhibits a conductance plateau close to $2e^2/h$ (REF.⁵⁸) (FIG. 6d).

Taken together, recent tunnelling spectroscopy measurements provide strong evidence for the presence of MZMs in semiconductor–superconductor heterostructures. The high quality of materials, as well as improved control in devices, opens the possibility of studying more exotic properties of MZMs.

Coulomb blockade experiments with proximitized nanowires. Improvements in the quality of interfaces in epitaxially grown proximitized nanowires open the possibility of studying the interplay of topological superconductivity and Coulomb blockade phenomena^{52,153}. The experimental device consists of a mesoscopic proximitized nanowire with a finite charging energy E_C that is coupled to two normal leads (FIG. 7a). In the Coulomb blockade regime, the charge on the island can be controlled by changing the gate voltage V_g as well as by electrostatically tuning single-channel barriers. Both these changes enable investigation of the crossover from strong to weak Coulomb blockade regimes and study of quantum charge fluctuations in the island. Quantum charge fluctuations have an imprint of the physical state of the system and thus allow identification of the topological phase¹³⁴.

As explained above, physical observables change their dependence on the dimensionless gate voltage $\mathcal{N}_g$ across the topological phase transition¹²⁹, and the charge periodicity changes from $2e$ to $1e$. Therefore, the two-terminal conductance through the island at zero bias is expected to exhibit a crossover between these two limits as a function of the magnetic field. Theory¹³³ predicts that the conductance is determined by different processes as the magnetic field is increased from zero, with a Cooper-pair-dominated regime for $B < B^*$, a sequential tunnelling regime for $B^* < B < B_c$ and a Majorana-dominated regime for $B > B_c$ (here, B^* is the field at which the ground state can have different parity, that is, $\Delta(B^*) = E_C$). For $B < B^*$, the ground state of the system involves only the even-parity sector. Therefore, the zero-bias conductance peaks appear at the degeneracy points between nearest even-charge states (FIG. 7b). For $B^* < B < B_c$, the ground state changes its parity between even-charge and odd-charge sectors as a function of $\mathcal{N}_g$, and zero-bias conductance peaks mark the change in ground-state parity. Thus, each conductance peak splits into two at $B = B^*$ (FIG. 7b), and this splitting increases with magnetic field until the field reaches B_c . At these degeneracy points, the dominant conduction mechanism is the resonant transfer of single electrons rather than Cooper pairs. In the intermediate regime, the peak conductance at low temperatures becomes suppressed (FIG. 7b) and, in fact, is expected to vanish in the thermodynamic limit where the wire length tends to infinity^{133,154}. Finally, in the topological regime ($B > B_c$), conductance peaks occur at half-integer values of $\mathcal{N}_g$, independent of the magnetic field (up to finite-size corrections). In this limit, the dominant contribution to the peak conductance originates from resonant tunnelling via MZMs. Owing to the non-local nature of the fermionic mode shared between two MZMs, the height of the

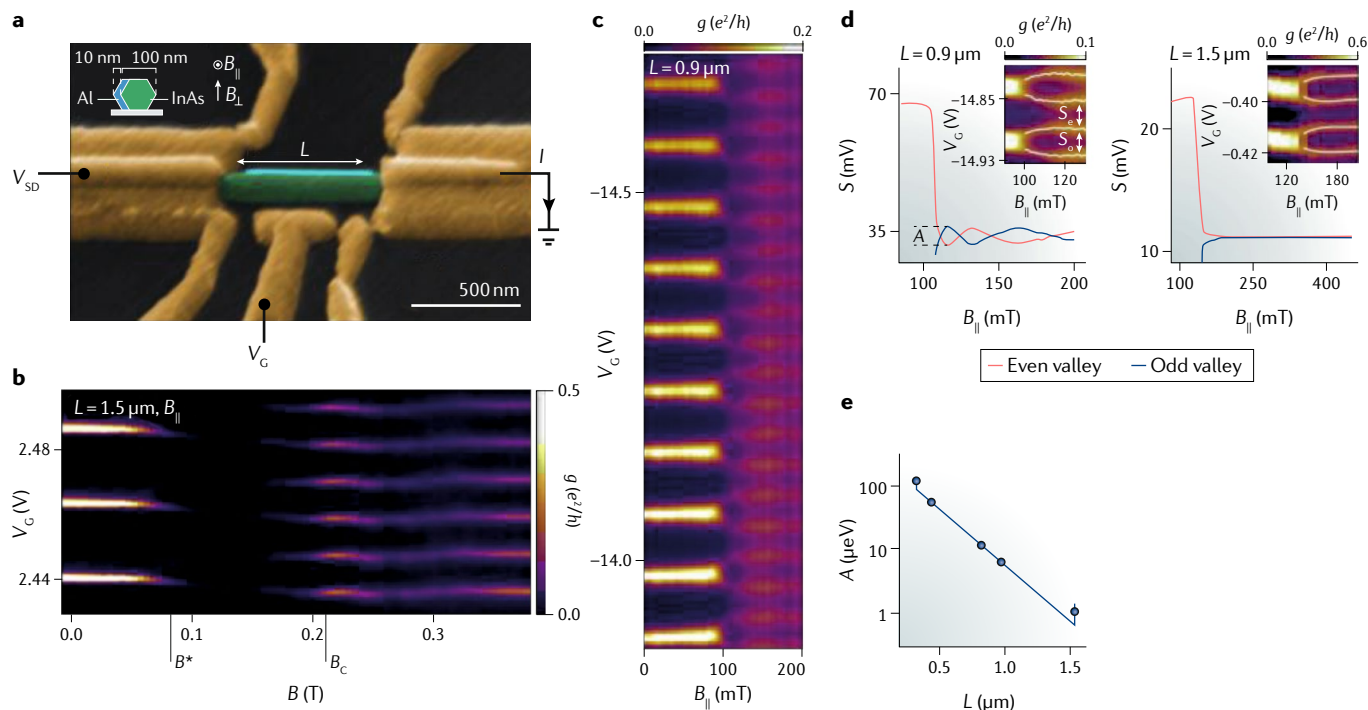


Fig. 7 | Coulomb blockade experiment with proximitized nanowires. **a** | Electron micrograph (false colour) of a device that is lithographically similar to the measured devices. The gold contacts are shown in yellow, the InAs nanowire is shown in green and the two-facet Al shell of length L is shown in light blue. Inset: cross section of a hexagonal InAs nanowire, showing the orientation of the parallel and perpendicular components of the magnetic field, $B_{\parallel}$ and $B_{\perp}$, respectively. **b** | Two-terminal conductance, g , as a function of an external parallel magnetic field and gate voltage for $L = 1.5 \mu\text{m}$. Coulomb peaks become dim at field B^* and brighten at field B_c . The field B^* is the characteristic scale for the crossover from the coherent Cooper pair to the incoherent single-electron tunnelling regime, whereas the field B_c corresponds to the topological phase transition. At $B > B_c$, transport is dominated by the coherent single-electron tunnelling processes owing to Majorana zero modes^{129,133}. **c** | Zero-bias conductance g as a function of gate voltage V_G and parallel magnetic field $B_{\parallel}$ for an $L = 0.9 \mu\text{m}$ device, showing a series of $2e$ -periodic Coulomb peaks below 100 mT and $1e$ -nearly periodic peaks above 100 mT. **d** | Average peak spacing for even and odd Coulomb valleys, S , as a function of magnetic field for $L = 0.9 \mu\text{m}$ and $L = 1.5 \mu\text{m}$. Even and odd peak spacings are indicated by arrows in the inset. **e** | Oscillatory amplitude, $A = S_e - S_o$, plotted against the proximitized nanowire length L for five devices, with L ranging from 330 nm to $1.5 \mu\text{m}$. The line is an exponential fit to $A = A_0 \exp(-L/\xi)$, with the parameter $A_0 = 300 \mu\text{eV}$ and the coherence length $\xi = 260 \text{ nm}$. Error bars indicate uncertainties propagated from lever arm measurements and fits to peak maxima. I , source–drain current; V_G , gate voltage; V_{SD} , source–drain voltage. Panels **a–e** are adapted from REF.⁵², Macmillan Publishers Limited.

conductance peak is much larger than when $B^* < B < B_c$ and is independent of the wire length.

The non-local nature of topological ground-state degeneracy can be investigated by studying the change in the spacing of the Coulomb blockade peaks, which originates from an exponentially small Majorana wavefunction overlap. The corresponding finite splitting energy $\delta E \propto \exp(-L/\xi)$ (where L is the wire length and ξ is the coherence length) leads to an energy difference for even-parity and odd-parity ground states and results in a small shift in the charge degeneracy points. Thus, the peak spacing within the Majorana scenario in the even-charge and odd-charge sectors is expected to be different, whereas in the normal-island case, this difference is negligibly small. In addition, the peak spacing depends on the nanowire length (FIG. 7c–e); thus, the experimental results⁵² are consistent with the MZM scenario. The splitting measurement provides important information about the physical system (such as superconducting coherence length), and it would be interesting to analyse this issue more systematically and to compare

experimental results with microscopic simulations of the proximitized nanowires¹⁵⁵.

Towards topological quantum computation

In the past 5 years, a new field of research into mesoscopic topological superconductivity has emerged^{52,123,129–131,133–135,137–139,150,156–165}, accompanied by an impressive growth in experimental explorations of various superconducting heterostructures^{46,48–51,53,55,84,98–100,103,153,166}. The interplay of mesoscopic and topological physics makes it possible to manipulate quantum information stored in topological (non-local) degrees of freedom in a way that minimizes decoherence effects. Remarkable experimental progress, rapid improvement in materials quality and an enhanced theoretical understanding of the microscopic details^{167–170} are encouraging developments for the prospect of building topological qubits in superconductor–semiconductor heterostructures.

In topological quantum computation, quantum gates rely on non-local transformations within the degenerate ground-state manifold, which can be performed either

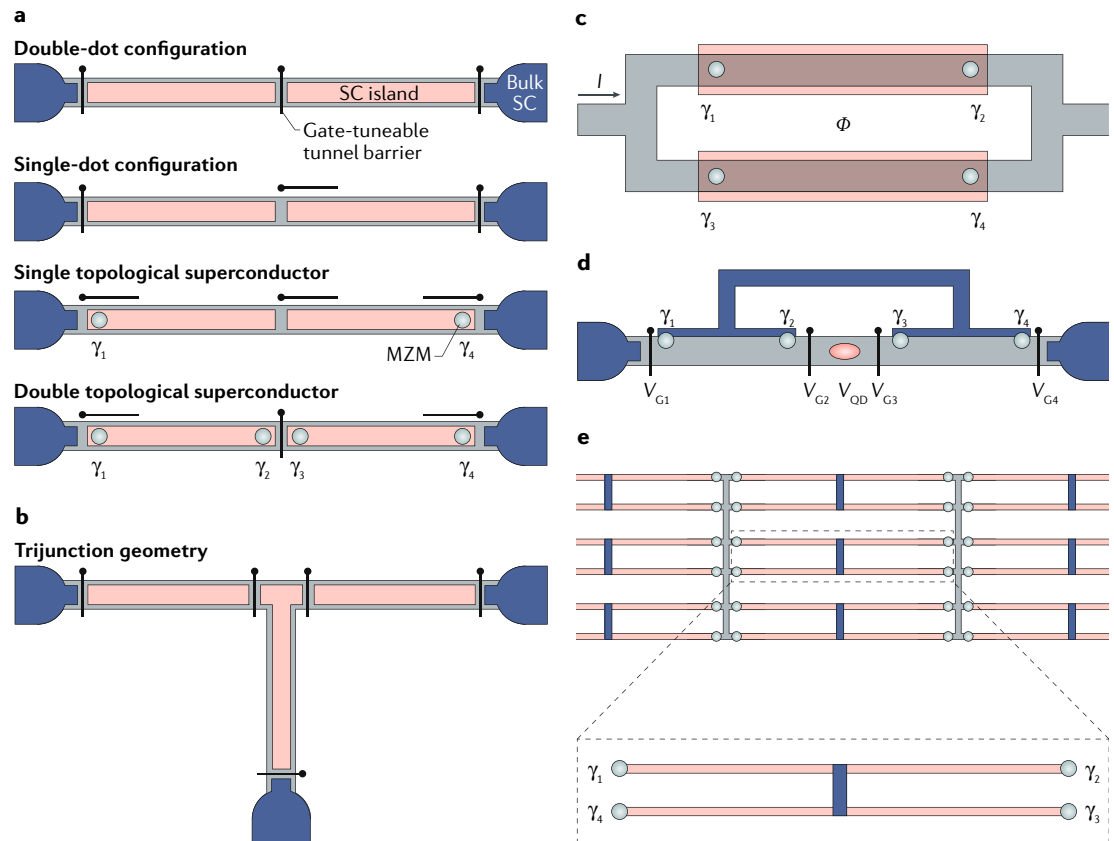


Fig. 8 | Proposed experiments to probe key properties of topological superconductors for quantum computing applications. **a** | The Majorana fusion experiment. The device consists of two mesoscopic Majorana islands (pink) with gate-tuneable tunnel barriers that act as valves. The charging energy of the superconductor (SC) island can be quenched by lowering the barrier to a bulk SC (blue). The middle barrier connects the islands. When the middle valve is completely open, the system effectively corresponds to a single island. Protocols of opening and closing the valves allow investigation of Majorana fusion rules. γ_{1-4} indicate the positions of the Majorana zero modes (MZMs). **b** | Topological trijunction (T-junction) for a braiding experiment. Anyonic state teleportation, which is equivalent to physically braiding MZMs, can be achieved by closing (and opening) the valves^{156,161}. **c** | A Majorana interferometer, which is a basic element of the Majorana-based surface code^{159,162,163}. Coherent transport through the interferometer probes the non-local nature of MZMs. **d** | Minimal setup for a quasiparticle-poisoning-protected topological qubit^{164,165}. The mesoscopic SC island hosting four MZMs is in a Coulomb blockade regime with fixed total charge. The island has twofold ground-state degeneracy and represents the simplest topological qubit. A semiconductor quantum dot, coupled to the qubit through barriers that can be controlled with gate voltages V_{Gi} , is used to perform projective measurements. Parity-dependent charge fluctuations between the semiconductor dot and SC island shift the energy levels in the quantum dot, making it possible to distinguish between different degenerate states of the qubit. The charge on the quantum dot can be tuned by changing the voltage V_{QD} . **e** | Scalable architecture for Majorana-based quantum computing. Topological qubits are coupled through semiconducting quantum dots. Quantum computation is performed using a measurement-only protocol, which is facilitated by tuneable couplings between MZMs and nearby semiconductor quantum dots¹⁷³. Φ , enclosed magnetic flux; I , current. Panels **a** and **b** are reproduced from REF.¹⁶¹, CC-BY-3.0. Panel **e** is reproduced with permission from REF.¹⁶⁵, American Physical Society.

by adiabatically changing the physical parameters of the system^{156,157,161,171,172} or by employing projective measurements^{164,165,173,174}. We now outline several immediate directions that should be explored both experimentally and theoretically in order to test the necessary ingredients for a successful operation of a future quantum computer.

The first step is to establish that the exotic non-local properties of MZMs are indeed realized in the experimental system. A hallmark of topological states of matter is ground-state degeneracy, which has not been tested experimentally to date. Fusion and braiding rules associated, respectively, with bringing together two initially

separated MZMs or exchanging their positions are characteristic of a topological phase and are of fundamental importance for topological quantum computing. Fusion and braiding experiments can be performed in a small network of nanowires (FIG. 8a,b). Fusion experiments are simpler than braiding experiments because they do not require an exchange of MZM position but nevertheless reveal topological properties of the system. Fusion rules can be tested using a device consisting of two Majorana islands connected together and to the superconducting leads by gate-tuneable tunnel barriers^{156,161} (valves, FIG. 8a). The effective charging energy of each island can be reduced by lowering the barrier connecting it to the

superconducting lead. We denote open (low barrier) and closed (high barrier) valves by '1' and '0', respectively. The string 010, for example, represents the configuration in which the left and right valves are closed and the middle valve is open. This configuration of valves effectively merges the left and right islands into a single island disconnected from the superconducting reservoirs. Two distinct sequences of open and closed valve configurations, differing only by one intermediate configuration, eventually lead to different final charge configurations on the islands¹⁶¹. This difference descends from the fusion rules of non-Abelian MZMs. While the sequence a_1 : 010 $\rightarrow$ 000 $\rightarrow$ 101 $\rightarrow$ 111 leads to a definite deterministic charge configuration on the left and right islands, the sequence a_2 : 010 $\rightarrow$ 111 $\rightarrow$ 101 $\rightarrow$ 111 results in an equal superposition of two distinct charge distributions. A measurement of the charge on the island following an execution of sequence a_1 should be deterministic. By contrast, after the execution of sequence a_2 , two charge configurations with equal probability are expected.

For braiding operations, three Majorana wires that form a T-junction (FIG. 8b) are necessary^{156,161}. The elementary sequence requires controlling five valves, that is, keeping the left-most and right-most valves open and tuning the remaining three valves¹⁶¹. Following the above notation, the sequence b_1 : 10001 $\rightarrow$ 11101 $\rightarrow$ 10111 $\rightarrow$ 10001 results in braiding of the two MZMs next to the operative left and right valves. Here, the numbering of valves goes from left to right, with the middle number representing the coupling of the middle island to the lead. The double braiding operation should always lead to a flip in the parity of the qubit, which is another non-trivial test.

At the heart of measurement-only topological quantum computation is a projective measurement of the joint fermion parity encoded in a pair of MZMs. In this scheme, an effective braiding between two non-Abelian anyons is achieved by measuring the topological charge of anyonic pairs (that is, fermion parity) rather than physically exchanging their position^{173,174}. Interferometric measurement is a promising way to distinguish between even and odd fermion parities; see, for example, the device shown in FIG. 8c. As explained above, transport through a proximitized nanowire in the Coulomb blockade regime changes its periodicity with the gate charge from $2e$ to $1e$ as a function of applied magnetic field. In the topological regime (large magnetic field), charge transport is dominated by coherent single-electron transmission through MZMs^{129,133,135}, which is very different from the inelastic transport in a normal metal island, although both systems exhibit a $1e$ periodic signal. Thus, the interferometric measurement should be able to distinguish between different tunnelling processes: transmission through MZMs or tunnelling through some other delocalized subgap state. Transmission through MZMs preserves electron coherence and therefore should depend on the enclosed magnetic flux Φ , whereas other processes do not. Although this setup cannot access properties of topological qubits (because the MZM parities are fixed by the charging energy), such an experiment is a crucial test of the concept of measuring MZM states using coherent links and

represents a basic building block of Majorana-based surface^{159,162,163} and colour¹⁷⁵ codes.

To perform quantum information processing, a mesoscopic superconducting island should support at least four MZMs that maintain ground-state degeneracy and have a fixed number of electrons owing to a large charging energy. The simplest topological qubit is depicted in FIG. 8d. The projective measurement can be performed by coupling MZMs to a central semiconductor quantum dot and measuring, for example, the quantum capacitance of the system. Indeed, virtual exchange of electrons between the semiconductor quantum dot and Majorana island results in a parity-dependent contribution to the ground-state energy, which ultimately leads to a change in the quantum dot charge distribution^{164,165,176}. In a more complex setup, Majorana islands supporting six MZMs open the possibility of performing a measurement-only braiding operation by the sequential measurement of the fermion parity of different pairs of MZMs¹⁶⁵.

Finally, an array of topological qubits coupled together through semiconducting quantum dots represents a scalable architecture for Majorana-based quantum computing¹⁶⁵ (FIG. 8e). Quantum computation is performed using the measurement-only protocol, which is facilitated by tuneable couplings between MZMs and nearby semiconductor quantum dots. A detailed protocol for implementation of Clifford gates using a limited set of projective measurements in this architecture has also been reported¹⁶⁵. Combining the topologically protected Clifford gates with the ability to produce and distil magic states^{177–183} should ultimately lead to universal quantum computation with MZMs.

Summary and outlook

The search for MZMs in superconductor–semiconductor heterostructures is evolving at a rapid pace and now encompasses a broad range of disciplines from materials science to condensed-matter physics and quantum information science. In less than a decade since the first theoretical proposals^{25,26}, compelling evidence for MZMs in proximitized nanowires has accumulated. Most of the experiments searching for MZMs have focused on tunnelling spectroscopy. Recent work on InSb/Al proximitized nanowires⁵⁸ reported robust $2e^2/h$ conductance, consistent with the presence of MZMs. The dependence of the zero-bias peak on the chemical potential and magnetic field has been measured, allowing the topological phase diagram to be mapped^{53,54}. In a different approach, experiments on floating proximitized nanowires⁵² set the stage for exploration of the Coulomb blockade effect in mesoscopic topological superconductors. The measurement of the 4π -periodic fractional Josephson effect was recently reported⁶². Although these experiments strongly support the existence of emergent MZMs in proximitized nanowires, direct observation of the non-Abelian properties associated with Majorana quasiparticles is still lacking. We outlined future experiments for the detection of more exotic properties of MZMs, such as braiding and fusion rules, which rely on observing two MZMs at the ends of a single wire simultaneously.

Experiments with proximitized nanowires not only contribute to our understanding of topological phases

of matter but also have significant theoretical and practical ramifications beyond condensed-matter physics. Next-generation experiments with Majorana nanowires will set the stage for the manipulation of quantum information in topological systems. Scalable designs for topological quantum computation based on superconducting islands hosting multiple pairs of MZMs have been put forward in the past 2 years^{164,165}.

Experimental demonstration of the exotic physics associated with MZMs, such as non-Abelian statistics,

would constitute a breakthrough for all of fundamental physics, as well as pave the way for the next milestone: validation of a topological qubit operating in the exponentially protected regime and exhibiting long coherence times¹⁸⁴. These accomplishments would lead to the exploration of different ideas for topological quantum memories and fault-tolerant universal quantum computation.

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Author contributions

All authors contributed to the discussion of content, researched data, wrote the article and assisted in editing the manuscript before submission.

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